

EMPLOYMENT

LexisNexis Legal & Professional

Senior Data Scientist I

Raleigh, NC

May 2024 - Present

- Technical lead for an agentic legal drafting system; led its migration from RAG and continue to own development.
- Reduced customer-facing product runtime by 24.8% through LLM call profiling and parallelization.
- Profiled high pod memory usage and introduced lazy module loading, reducing time at peak memory from 86.4% to 68% of runtime.
- On a 3-person team, increased the share of drafts rated suitable for users by attorney reviewers from 40% to 86% for complaints and 59.2% to 76% for motions.
- Built a contrastive-learning filter to remove unrelated legal passages from search results; median filtering latency of 12ms across 40 queries.
- Mentored an intern on a large-scale, full factorial experiment investigating Pareto-optimal configurations for draft quality and latency.
- Built team developer tooling: a Rust MITM proxy with request replay, a web UI for visualizing intermediate agent steps, and custom Python lint rules beyond `ruff`'s rule set.
- Technology:** Python, Litestar, Google ADK, React, Rust, TypeScript, Tailwind

North Carolina State University

PhD Student

Raleigh, NC

Aug 2019 - Jul 2024

- State-of-the-art hyper-parameter optimization:** Proposed a novel hyper-parameter optimization method that outperforms prior SOTA and is 36.4% faster.
- Better, faster deep learning for SE:** Improved defect prediction by up to 123% (F-1 score), code smell detection by up to 30% (AUC), issue lifetime prediction by up to 76% (accuracy), automated microservice partitioning by up to 285% (modularity) compared to prior SOTA.
- Semi-supervised learning:** Achieved state-of-the-art results (up to 100% improvement in AUC) on static code warnings analysis using 10% of the labels.
- Teaching:** TA for 830 students over 5 semesters across CSC 230 (C and Software Tools), CSC 510 (Software Engineering), and CSC 591/791 (Automated Software Engineering)

RELEVANT PROJECTS

pi-permission-system (maintained fork)

TypeScript, Node.js

May 2026 - Present

[GitHub](#)

Maintain a permission-system fork for the Pi coding agent. Redesigned shell permissions using grammar-based analysis of pipelines, substitutions, and redirections, with protected-path checks and argument-aware approval rules. Extended enforcement to background commands and fixed concurrent subagent approval handling.

Q&A System for Zotero

Rust, SQLite, LanceDB

Oct 2024 - Present

[GitHub](#) :: [crates.io](#)

Built an academic question-answering and semantic search system over a user's Zotero library. Wrote a custom PDF parser for academic papers (median throughput $\approx$ 2,400 pages/s across 11 PDFs; M4 Pro) and a RAG library with support for multiple providers and tool calling.

RAISE

Python, Keras

Aug 2020 - Jun 2025

[GitHub](#) :: [PyPI](#)

Sole developer of an ML utilities Python package used by multiple research groups for reproducible experiments. Downloaded 74k+ times on PyPI.

Google/Meta Data Mining

Python, Keras

Feb 2021 - May 2021

[GitHub](#)

Mined Google Takeout and Meta data exports to predict which Amazon best-sellers to advertise to a user (DistilGPT-2, 0.6 F-1 score).

Novel Drug Repurposing Hypotheses

Python, PyTorch

Oct 2019 - Feb 2020

[GitHub](#)

Identified novel drug repurposing hypotheses by text mining transcripts of an NPR medical radio show, and verified results using a knowledge graph.

SKILLS

Languages: Rust, Python, TypeScript, Java, C++, Gleam
ML/AI: Keras, PyTorch, Hyper-parameter optimization, RAG, LLM evaluation, Agentic systems, Google ADK
Frameworks: React, Flask, Litestar, FastAPI
Databases: LanceDB, MySQL, MongoDB, DynamoDB
Cloud: Google Cloud (Compute Engine, Cloud Storage, Vertex AI), AWS (EC2, S3, DynamoDB)

EDUCATION

North Carolina State University <i>Ph.D. Computer Science</i> ◦ Advisor: <i>Dr. Tim Menzies</i> ◦ Dissertation: <i>Guidelines for the Application of Neural Technologies in Software Analytics (or: How to Do More with Less in SE)</i>	Raleigh, NC <i>Aug 2019 - Jul 2024</i>
PES University <i>B.E. Computer Science – Advisor: Dr. Snehanshu Saha</i>	Bangalore, India <i>Aug 2015 - May 2019</i>

HONORS

Distinguished Reviewer, ICSE 2026	Mar 2026
Google Developer Expert - ML and GCP	Mar 2025 - Present
Google Cloud Champion Innovator - Cloud AI/ML	Oct 2022 - Mar 2025
Google Cloud Research Innovator	Feb 2022

RECENT PUBLICATIONS

See full list on Google Scholar.

1. **Yedida, R.**, & Menzies, T. (2025). Is Hyper-Parameter Optimization Different for Software Analytics? *IEEE Transactions on Software Engineering*, 51(6).
2. Baldassarre, M. T., Ernst, N., Hermann, B., Menzies, T., & **Yedida, R.** (2023). (Re)use of Research Results (is Rampant). *Communications of the ACM*, 66(2), 75-81.
3. **Yedida, R.**, Kang, H. J., Tu, K., Lo, D., & Menzies, T. (2023). How to Find Actionable Static Analysis Warnings: A Case Study with FindBugs. *IEEE Transactions on Software Engineering*, 49(4), 2856-2872.
4. **Yedida, R.**, Krishna, R., Kalia, A., Menzies, T., Xiao, J., & Vukovic, M. (2023). An Expert System for Redesigning Software for Cloud Applications. *Expert Systems with Applications*, 219, 119673.
5. **Yedida, R.**, Menzies, T. (2022). How to Improve Deep Learning for Software Analytics (a case study with code smell detection). In *2022 IEEE/ACM 19th International Conference on Mining Software Repositories (MSR)*.